Measurement-Free Error Correction in Rydberg atom array

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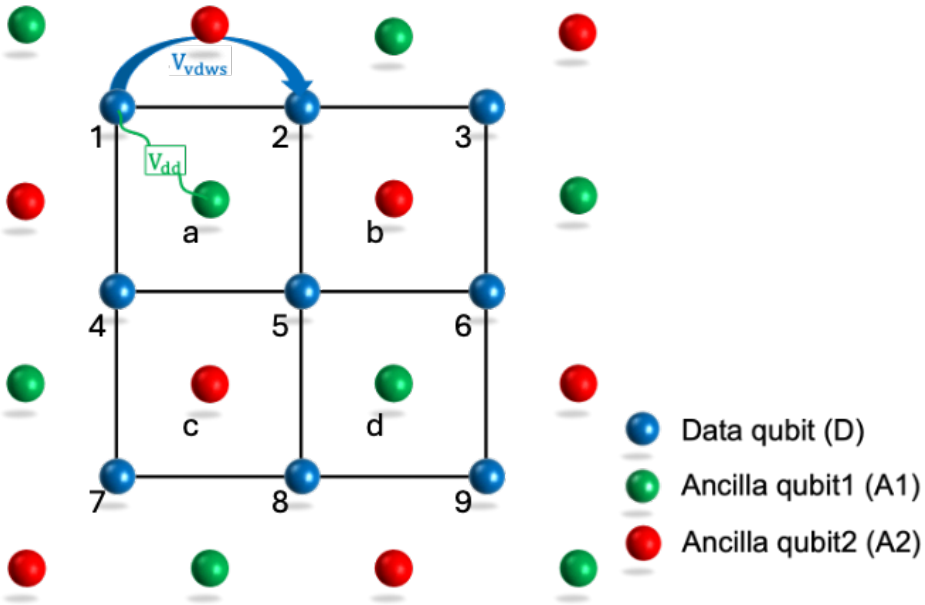
2025-07-28

Motivation

- Developing a one-step parity check process with global pulse sequence in Rydberg atom system without multi-step movements of atoms.
- The errored qubit can be corrected by the ancilla qubits without measurement (Ancilla qubits acts as control qubits and errored data qubit as target qubit.[1])

Model

- Three different species.(Blue→data qubits; red/green→ancilla qubit A2/A1)
- Parity check: (Z-error in plaquette A2), (X-error in plaquette A1)
- Strong interaction between different species, weak interaction between same species.



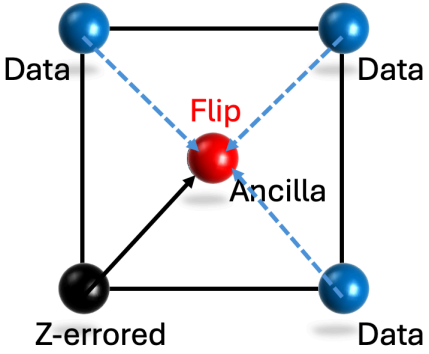
Syndrome extraction can be done with parity-check process

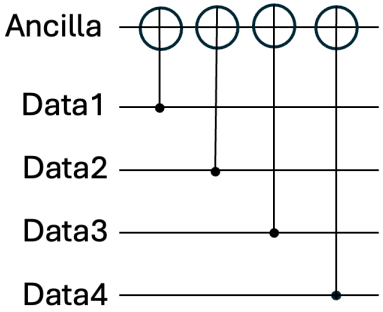
- We intend to perform parity-check process (with the ancilla qubit acts as target qubit and the neighboring data qubit acts as control qubit)

– no error → all qubits keep invariant after the effective pulse sequence.

– X-error → neighboring ancilla qubit A1(green dots) will flip.

– Z-error → neighboring ancilla qubit A2(red dots) will flip.





Parity check process is hard to design with global pulse sequence

- The one-step parity check process is not easy to implement:

– We intend that only when the number of errored data qubits is odd, the neighboring ancilla qubit A1 will flip.

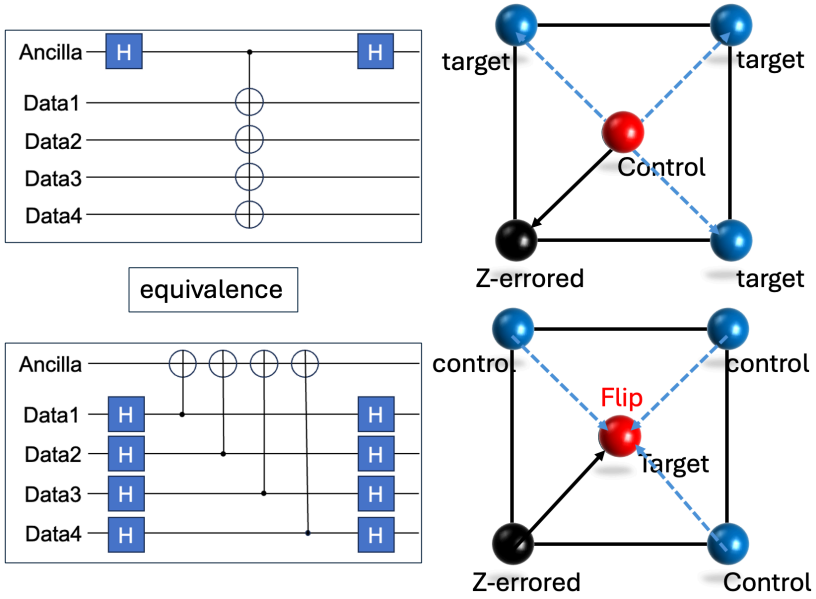
- In Rydberg atom system:

– Parity check under blockade effect.

– Parity check under interaction-conpensation scheme(anti-blockade effect).

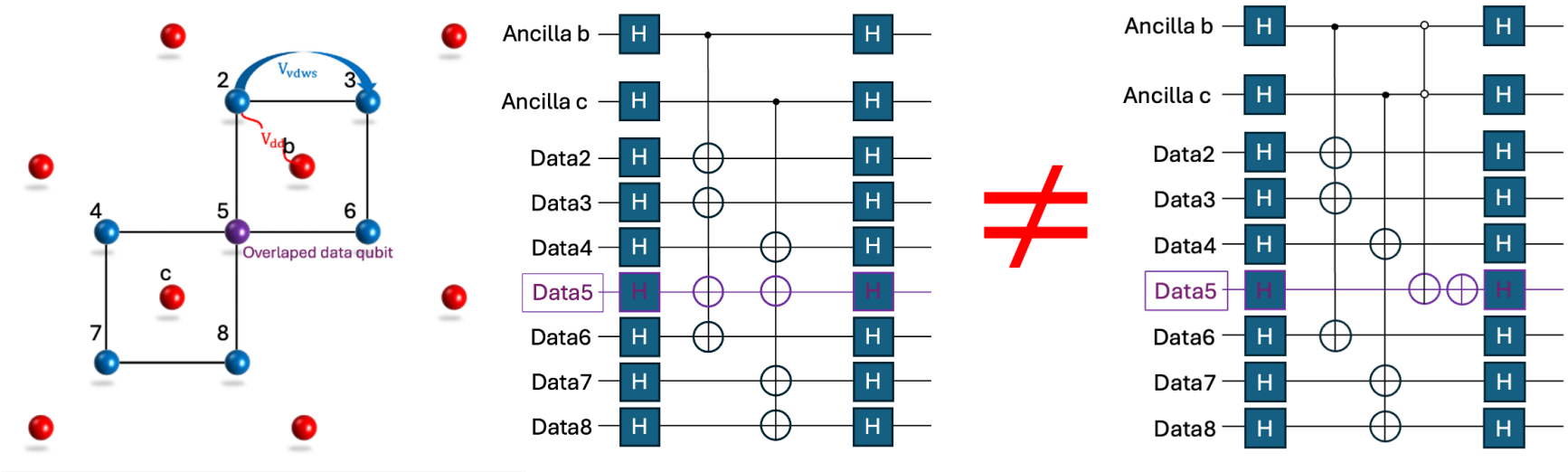
Parity check can be implemented equivalently on gate level [2]

- In one plaquette, we can use hardmard gates to exchange the control and target qubits.
- The effective parity check can be implemented with ancilla qubit as control qubit and data qubit as target qubit.



Global pulse implements different circuit

- Overlapped data qubits:



- For global pulse sequence, we need to change the gate scheme mentioned above ⇒ implement the operation shown in left-hand circuit through one global pulse.

Main problem:

- Hard to construct one-step multi-qubit parity gate for perfect parity check.

  $\Rightarrow$ We construct effective gate for parity check

- Because of the interactions between data qubits and ancilla qubits, I cannot determine the minimum unit for gate-level pulse design.

$\Rightarrow$ We simulate to verify the effectiveness of the designed pulse sequence.

- feasibility of the designed gate scheme in large system(experimental perspective)(for example, more than 2 species).

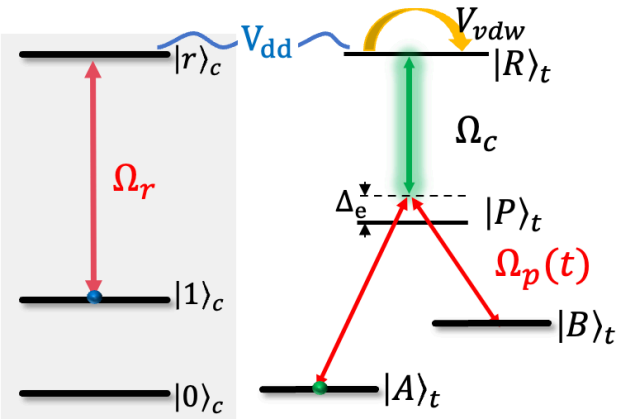
Appendix: Multi-qubit CNOT gate

- In order to perform the CNOT_4 gate, we intend to use Rydberg blockade effect and:

– When ancilla qubit is in state $|0\rangle$, the data qubits are unchanged.

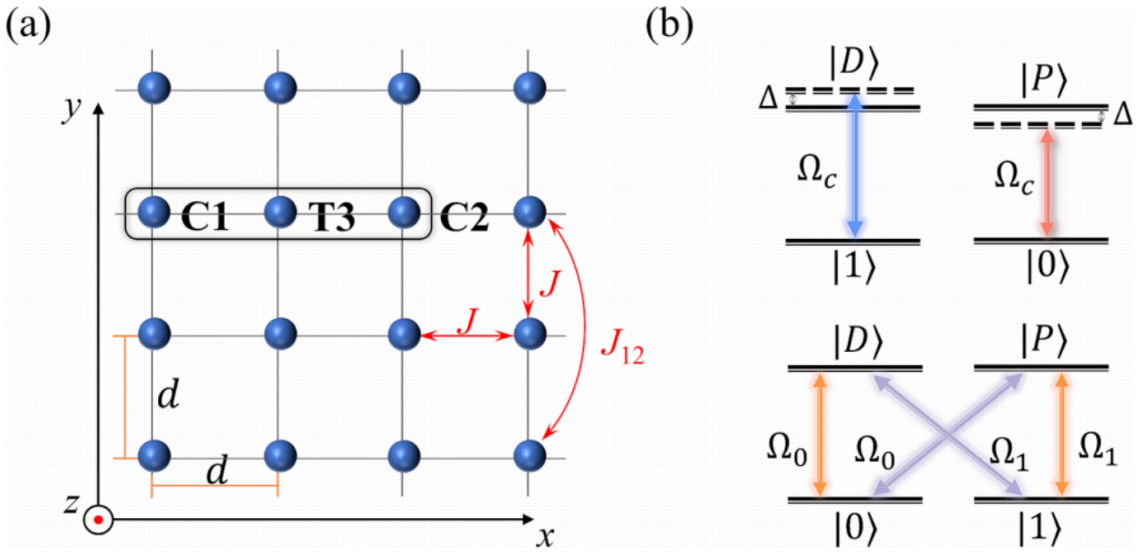
– When ancilla qubit is in state $|1\rangle$, the data qubits are flipped.

- For example:



Appendix: Parity Gate scheme

- The overlapped data qubit requires that when both neighboring ancilla qubits are in state $|1\rangle$ or $|0\rangle$, the target data qubit will be unchanged.[3]



- The scheme is implemented through anti-blockade effect which requires accurate control of the interaction strength between data qubits and ancilla qubits.

Appendix: Parity Gate scheme

- Double control considered(whether the parity gate scheme can achieve effective parity check):

$$|0_c 0_c 0_t\rangle \leftrightarrow |0_c 0_c 0_t\rangle$$

$$|0_c 1_c 0_t\rangle \leftrightarrow |0_c 1_c 1_t\rangle$$

$$|1_c 0_c 0_t\rangle \leftrightarrow |1_c 0_c 1_t\rangle$$

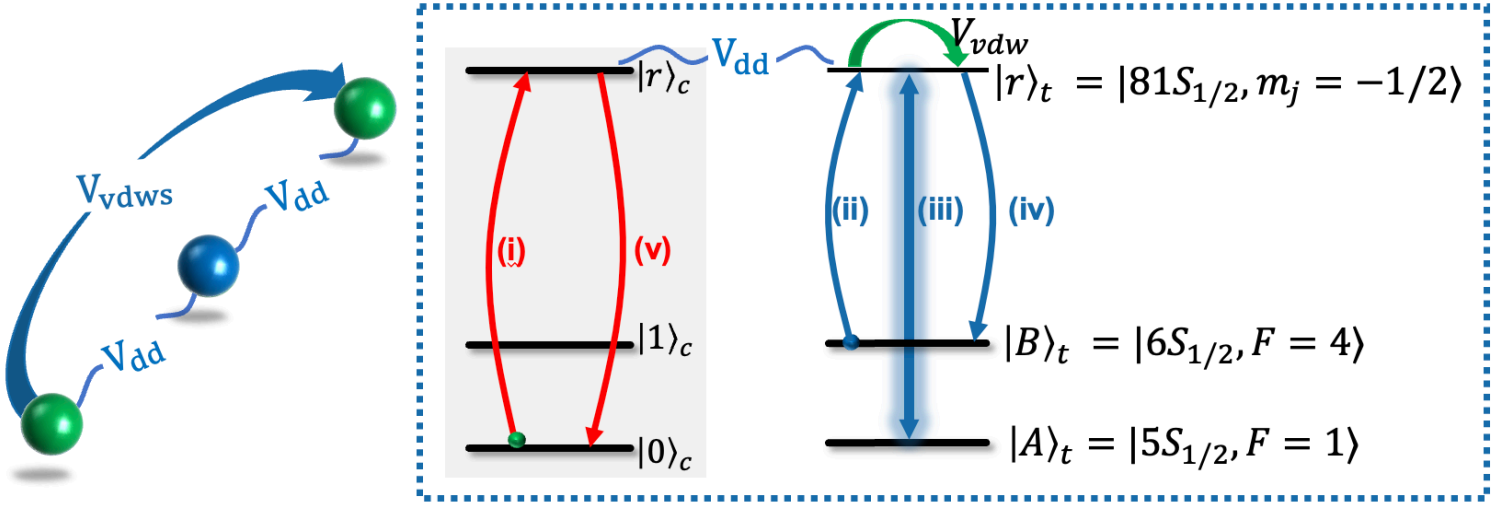
$$|1_c 1_c 0_t\rangle \leftrightarrow |1_c 1_c 0_t\rangle$$

Appendix: Error correcting C_2NOT gate

- For example, if there is a Z-error happens on data qubit 5, the parity check will flip the neighboring red ancilla qubits b,c. Then, we can perform a C_2NOT gate for error correction where the two flipped neighboring ancilla qubits act as control qubits and the errored data qubit acts as target qubit.
- Rule: Only if the ancilla qubits are both in state $|1\rangle$, the target data qubit will be flipped.

Appendix: Error correcting C_2NOT gate

- gate scheme:



- The C_2NOT gate can be constructed through the EIT effect and the Rydberg blockade effect.

Literature Review

1. Construction of multi-qubit CNOT_k gate through EIT:[2], through optimization of the pulse[4], through adiabatic pulse design[5], [6] and experimental realization:[7], [8]; Parity gate construction through specially designed interaction strength[3].
2. Measurement-free fault-tolerant quantum error correction:[1]
3. Mid-circuit correction of correlated phase errors using an array of spectator qubits[9]
4. mesoscopic Rydberg gate based on EIT:[5]. Experimental realization:[7], [8]
5. There is a scheme familiar to this work in one plaquette.[6]

Bibliography

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